

SCHOOL OF COMPUTER SCIENCE AND IT YBN UNIVERSITY RANCHI

JAVA PROGRAMMING ASSIGNMENT 09

Problem 1: Calculate the Area of a Circle

Description:

Write a Java program with a method named `calculateArea` that takes a single parameter, the radius of a circle (a double), and returns the area of the circle. The program should prompt the user to input the radius, call the method, and then print the calculated area.

Concepts Covered:

- Creating methods
- Using method parameters
- Returning values from a method

Hint:

- The formula for the area of a circle is $\pi * \text{radius} * \text{radius}$.
- Use `Math.PI` for the value of π .

Problem 2: Check if a Number is Prime

Description:

Write a Java program that includes a method named `isPrime` that takes an integer as a parameter and returns a boolean indicating whether the number is prime. The program should prompt the user to input a number, call the method, and print whether the number is prime.

Concepts Covered:

- Creating methods with parameters
- Returning boolean values from a method
- Using loops and conditional statements inside a method

Hint:

- A prime number is a number greater than 1 that has no divisors other than 1 and itself.
- Use a loop to check divisibility and return `false` if the number has a divisor.

Problem 3: Find the Maximum of Three Numbers

Description:

Write a Java program with a method named `findMaximum` that takes three integer parameters and returns the largest of the three numbers. The program should prompt the user to input three numbers, call the method, and print the maximum number.

Concepts Covered:

- Creating methods with multiple parameters
- Using `if-else` statements to compare values
- Returning values from a method

Hint:

- Use nested `if-else` statements or the `Math.max()` method to find the largest of the three numbers.